

From Disaster to Design

How Recent Flooding Changed Students' Views on Engineering in Nepal

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A student-led exploratory study

Abstract

This exploratory study examines how recent flooding in Nepal relates to students' self-reported interest in engineering and their views of the role engineers can play in reducing disaster impacts. An anonymous online survey was distributed through Google Forms and received 31 responses from students reached through personal and student networks. The survey included questions about engineering interest, students' earlier perceptions of engineering, priorities for reducing future disaster impacts, and open-ended responses. Twenty respondents (64.5%) reported that their interest in engineering had increased either significantly or somewhat. Respondents most often identified urban/community planning, roads and bridges, flood-resistant infrastructure, and early-warning systems as important areas for future engineering. Written responses also repeatedly mentioned infrastructure quality and maintenance, safety, planning, resilience, and drainage. Because the sample was small and voluntary, the findings are descriptive of this group and should not be generalized to all students in Nepal.

1. Why I Chose This Topic

Recent flooding made me think more seriously about the role of engineering in protecting people and communities. I had usually thought of civil engineering mainly in terms of buildings, roads, bridges, and other physical structures. Seeing how quickly flooding can affect those structures made me wonder whether other students were thinking about engineering differently as well.

I therefore decided to conduct a small survey to explore whether recent flooding was associated with a change in students' interest in engineering and what students thought engineers should prioritize when designing for future disasters.

2. Research Questions

- How did exposure to recent flooding relate to students' self-reported interest in engineering?
- What did students associate with engineering before the recent flooding?

- Which engineering priorities did students identify as important for reducing future disaster impacts?
- What ideas appeared repeatedly in students' written responses?

3. Methodology

3.1 Survey Design and Data Collection

I designed an anonymous online questionnaire using Google Forms. The survey link was distributed to students through existing personal and student networks, and participation was voluntary. No names were requested as part of the survey. A total of 31 responses were collected.

The questionnaire included a retrospective 1–5 rating of engineering interest before the recent flood events, questions about what engineering meant to respondents, questions about their exposure to the flooding, questions about engineering priorities for reducing future disaster impacts, and open-ended questions that allowed respondents to explain their views in their own words.

3.2 Data Analysis

I organized the responses in a spreadsheet and summarized closed-ended questions using counts and percentages. For questions that allowed multiple selections, each selected category was counted once per respondent, so totals could exceed 31. For the written responses, I read the answers and grouped recurring ideas into broad themes such as infrastructure maintenance, planning, safety, resilience, drainage, and site assessment.

3.3 Sampling

This study used a voluntary convenience sample rather than a statistically random sample. The survey was distributed to students who could be reached through available networks rather than to students randomly selected from a complete list of students in Nepal. This approach was practical for a student-led exploratory project, but it means that the 31 responses should not be treated as representative of students in Nepal as a whole.

4. Results

4.1 Change in Interest in Engineering

Twenty of the 31 respondents (64.5%) reported that their interest in engineering had increased after seeing or learning about the flooding. Ten reported an increase significantly and ten reported an increase somewhat. Six said their interest stayed about the same, four were still unsure, and one reported a decrease.

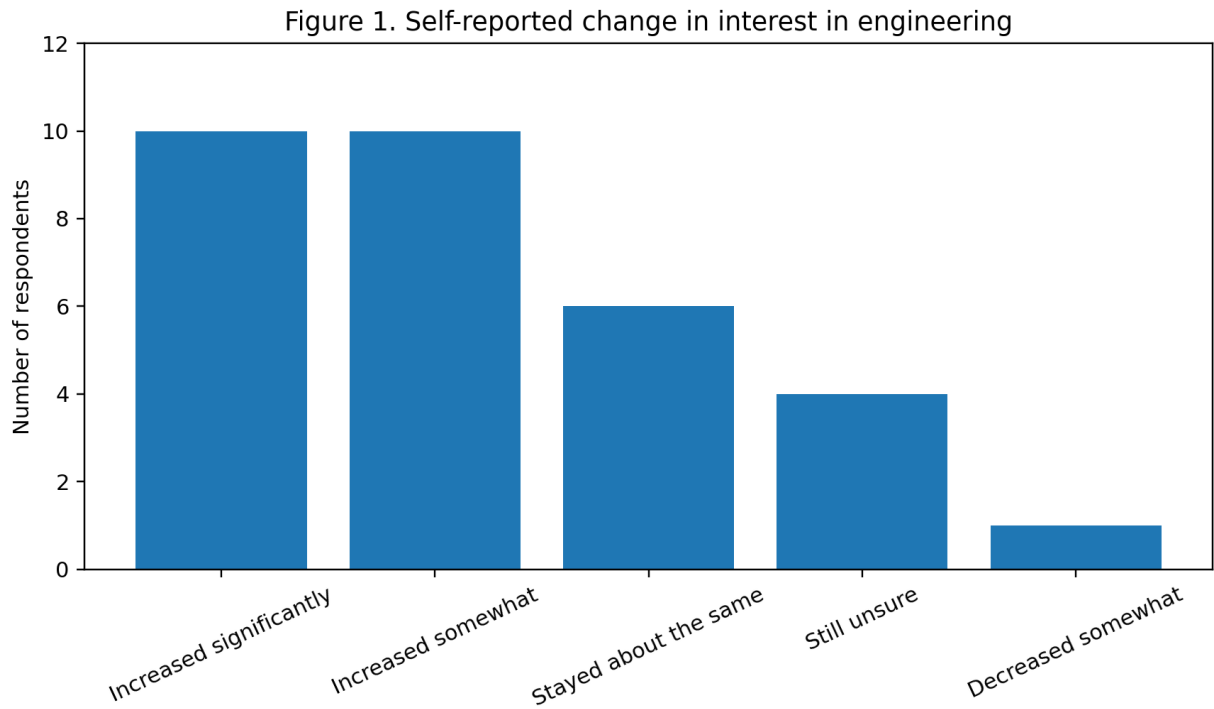


Figure 1. Self-reported change in interest in engineering (n = 31).

This result describes what respondents reported; it does not establish that the flooding caused the change in interest.

4.2 What Students Associated with Engineering

Before the flooding, buildings and construction was the most common association with engineering, selected by 18 respondents. Technology and innovation was selected by 12, while roads and bridges and machines and robotics were each selected by 11.

Figure 2. What respondents associated with engineering before the floods

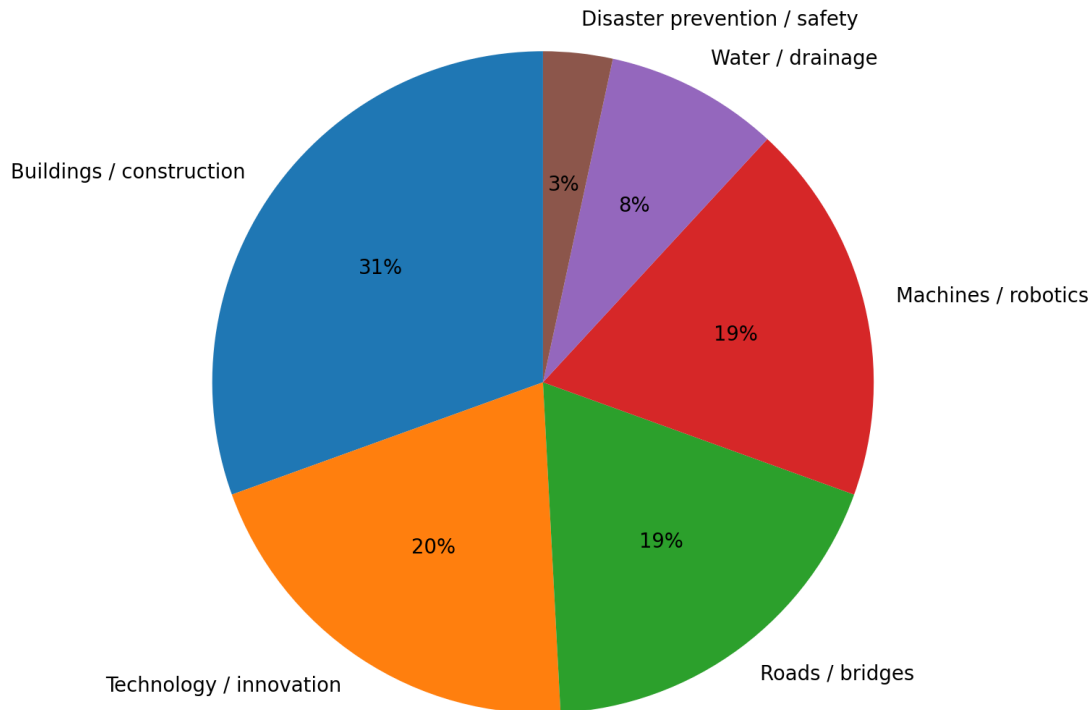


Figure 2. Respondents' selected associations with engineering before the recent flooding. This was a multiple-selection question, so percentages do not sum to 100%.

4.3 What Should Engineers Prioritize?

When respondents were asked which areas engineers should prioritize to reduce future disaster impacts, urban/community planning was selected by 15 respondents. Roads and bridges and flood-resistant infrastructure were each selected by 13, followed by early-warning systems and technology (12).

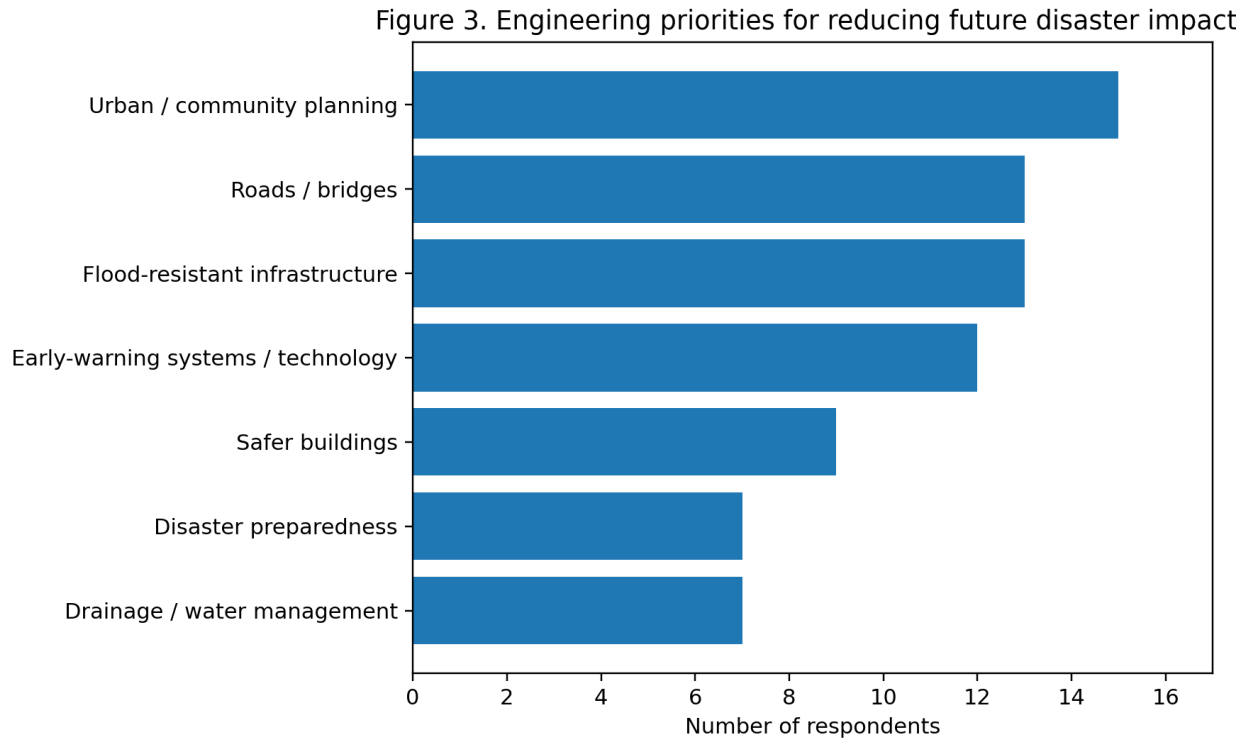


Figure 3. Engineering priorities identified by respondents.

5. What I Learned from the Written Responses

The open-ended responses helped me understand the numbers better. Several ideas appeared repeatedly, and many respondents described engineering as more than simply constructing stronger buildings or roads.

Recurring theme	Responses
Infrastructure quality and maintenance	26 (83.9%)
Planning, land use, and urbanization	15 (48.4%)
Safety and protection of life	14 (45.2%)
Resilience and disaster resistance	14 (45.2%)
Drainage and water management	13 (41.9%)
Sustainability	9 (29.0%)
Geotechnical or site assessment	7 (22.6%)
Climate adaptation or future risk	5 (16.1%)
Early warning or monitoring	4 (12.9%)

These themes overlap, so they are descriptive rather than independent statistical categories. Respondents discussed drainage, unplanned construction, settlement near rivers, site assessment, maintenance, and designing for extreme conditions.

6. What This Made Me Think About as a Future Engineer

Before doing this project, I mostly thought about civil engineering in terms of physical structures. The survey made me think more about the decisions behind those structures.

A road can be strong, but it can still be vulnerable if it is built in a risky location. A building can be well constructed, but planning and site selection still matter if it is placed near a river or unstable slope. Communities may also need drainage, monitoring, warning systems, and preparation in addition to physical infrastructure.

7. From Survey to Engineering Prototype

I did not want the project to end with collecting opinions. One result that interested me was that 12 respondents selected early-warning systems and technology as an important area for engineers. I therefore developed a simple student-level flood early-warning prototype as an engineering extension of the survey.

The prototype models a basic hazard-monitoring chain: a simulated water-level measurement is collected, compared with warning thresholds, and used to display a status and a changing graph. The current simulation uses three states—SAFE below 40 cm, WARNING from 40 cm to below 60 cm, and DANGER at 60 cm or above. These values are demonstration thresholds, not field-calibrated safety limits. The prototype is intended for learning and concept demonstration, not real emergency use.

The basic idea is to model a chain in which a water-level or other hazard measurement is collected, compared with a warning threshold, and used to trigger an alert. I would treat this as a prototype or simulation rather than a real emergency system. The purpose would be to learn how an engineering idea can move from a problem identified by people to a measurable and testable design.

8. Limitations

- The sample was small, with only 31 respondents.
- Participation was voluntary and the survey used a convenience sample, so the findings cannot be generalized to all students in Nepal.
- The survey asked respondents to remember their earlier level of interest in engineering, so those retrospective answers may not perfectly represent how they felt at the time.
- The study did not collect a controlled pre-flood baseline, so it cannot establish that the flooding caused the reported changes in interest.
- The open-ended responses were grouped into themes as a first-pass student analysis rather than through a formal independent coding process.

- The survey measures perceptions and opinions; it does not test whether a particular engineering solution would actually prevent flood damage.

For these reasons, I consider this project an exploratory study. Its purpose is to identify patterns in a small group of responses and use those patterns to guide further learning, rather than to make broad claims about Nepal's student population.

9. Conclusion

I started this project because recent flooding made me think about how engineers can help protect people in Nepal. After collecting 31 anonymous survey responses, I found that 20 respondents reported increased interest in engineering. Respondents also frequently identified planning, roads and bridges, flood-resistant infrastructure, and early-warning technology as important areas for future engineering.

The survey therefore became a starting point rather than the final product. Building the early-warning prototype allowed me to move from studying what students think engineers should do to exploring one engineering solution myself. The prototype remains a student simulation, but it gives the project a practical next step: connecting a community-level problem, survey evidence, and a measurable engineering concept.

The survey therefore became a starting point rather than the final product. My next goal is to build and test a small early-warning prototype so that I can move from studying what students think engineers should do to exploring one engineering solution myself.

Appendix A: Data at a Glance

Finding	Result
Survey responses	31
Reported increased engineering interest	20 (64.5%)
Urban/community planning selected	15
Roads and bridges selected	13
Flood-resistant infrastructure selected	13
Early-warning systems/technology selected	12
Safer buildings selected	9
Disaster preparedness selected	7
Drainage/water management selected	7

Appendix B: How the Analysis Was Done

The main tools used were Google Forms for data collection and a spreadsheet for organizing and analyzing the responses. Basic counts and percentages were used for the closed-ended questions. For example:

$$20 \div 31 \times 100 = 64.5\%$$

For the written responses, I read the answers and identified ideas that appeared repeatedly, then grouped similar ideas into themes. The purpose was to make the main patterns easier to see without presenting the qualitative analysis as more precise than it was.

Appendix C: Prototype and Source Code

The flood early-warning prototype was developed as a small HTML/CSS/JavaScript simulation. The full source code is provided as a separate project file and will be linked from the project website so that the implementation can be viewed without making the research paper unnecessarily long.

Core prototype logic:

- 1. Read a simulated water-level value from the interface.
- 2. Compare the value with the demonstration thresholds.
- 3. Display SAFE, WARNING, or DANGER.
- 4. Record the simulated value and display its change on a simple graph.
- 5. Make clear that the system uses simulated data and is not an operational flood-warning system.

The source code should be treated as part of the project artifact rather than as evidence for the survey findings. The survey data and the prototype serve different purposes: the survey provides exploratory evidence about respondents' perceptions, while the prototype demonstrates how one identified priority can be translated into a simple engineering concept.